

CLASS-8

LESSON-11 DIRECT AND INVERSE PROPORTIONS

(This PDF Based on NCERT Book)

DIRECT PROPORTION(प्रत्यक्ष अनुपात)-

Direct proportion describes a relationship where two things change in the exact same direction and at the exact same rate.

How it Works

Imagine you are buying apples. If 1 apple costs 1, then 2 apples will cost 2. If you buy 10 times as many apples, you pay 10 times as much money.

- **Double one?** The other doubles.
- **Halve one?** The other halves.
- **The Ratio Stays Constant:** If you divide one by the other, you always get the same number (in this case, the price per apple).

Real-World Examples

Situation	Direct Proportion Relationship
Driving	The longer you drive at a steady speed, the further you travel.
Working	If you get paid by the hour, the more hours you work, the bigger your paycheck.
Cooking	If you double a recipe, you need to double every single ingredient.
Shopping	The more kilograms of fruit you buy, the higher the total cost.

The "Math" Side

In mathematics, we say that y is directly proportional to x if they follow this formula:

$$y = kx$$

- **y and x** are the two things changing.
- **k** is the "constant"—the number that connects them (like the price of a single apple).

Example: Weighing Water

Water has a very simple relationship: **1 ml of water weighs exactly 1 gram.**

- **The Scenario:** You are filling a beaker with water.
- **The Variables:**
 - y = The Mass (grams)
 - x = The Volume (milliliters)
 - k = The Density (which is 1 for water)

The Formula: $y = 1x$

INVERSE PROPORTION(प्रतिलोम अनुपात)-

If **direct proportion** is about moving together, **inverse proportion** is about moving in opposite directions.

In simple words: **When one value goes up, the other value goes down.**

Real-World Examples

Situation	The Inverse Relationship
Speed vs. Time	The faster you drive, the less time it takes to get there.
Workers vs. Task	The more people helping to paint a fence, the less time it takes to finish.
Volume vs. Pressure	If you squeeze a balloon (decrease volume), the air pressure inside goes up .
Sharing Pizza	The more friends you share a pizza with, the smaller the slice each person gets.

The Formula: $y = \frac{k}{x}$

In math, we say y is inversely proportional to x using this formula:

$$y = \frac{k}{x} \text{ or } x \times y = k$$

- **x and y** are your two changing values.

- **k** is the "magic number" (constant) that stays the same.

A Simple Calculation Example

Scenario: It takes **4 hours** for **1 person** to clean a house. How long does it take **2 people**?

1. **Find k:** Multiply the values you know. 1 person $\times$ 4 hours = 4. So, **k = 4**.
2. **Use the formula:** $y = \frac{4}{x}$
3. **Plug in the new number:** If x (people) is 2, then $y = \frac{4}{2}$.
4. **Result:** It takes **2 hours**.